Group Q – Earlier Self-Assessment Design Notes

Historical design record

This document retains the earlier Assess, Check & Learn and Query design. The project direction described by the user on 12 October 2026 focuses on research similarity and distinctiveness; these earlier modules are not the current agreed scope. Four-member confirmation of the revised design is recorded separately in the project alignment board.

1 Phase 2 Mock-up: Screen Walkthrough and Design Rationale

1.1 Home screen: choose a module

The prototype opens with three menu choices. The walkthrough below sketches each route; the Check & Learn screen is a proposed interpretation of its module description for the responsible teammate to review. All records and figures shown in the walkthrough are fictional demonstration data, not measured TU/e results.

Home — AI for a Greener Research Future

What would you like to do?

1. I want to Assess | 2. I want to Check & Learn | 3. I want to Query

Selected route: Option 1 opens Section 1.2; Option 2 opens Section 1.3; Option 3 opens Section 1.4. A researcher can return here from any route.

1.2 Option 1 — I want to Assess: a guided project assessment

This walkthrough uses a laboratory battery-sample project to show where fixed controls, AI assistance and a short conversation each help. The same sequence could ask different activity-specific questions for GPU computation or a printed prototype. The tool proposes an activity type and follow-up prompts; the researcher confirms them. Arithmetic uses recorded quantities, not an AI-generated score.

A. Start from the menu and confirm the activity

Home > 1. I want to Assess > New assessment

Choose a starting point: Battery or laboratory samples | GPU computation | 3D-printed prototype | Describe another activity

Researcher selects: Describe another activity.

Researcher

I make lithium-ion battery electrode samples. I want to reduce wasted material and electricity without changing which samples count as scientifically valid.

AI-assisted draft: This appears to be a laboratory battery-sample activity. I would assess one batch and one valid sample, while retaining total batch use. Please confirm the activity and define a valid sample.

Confirm battery-sample activity | Edit description | Choose another activity

B. Fill known fields; ask only for a missing condition

Project information form — illustrative prior batch

Scientific outcome	Electrode samples meeting the same test requirement.
Prior batch	6 attempts; 5 valid samples.
Relevant inputs	60 kWh for the defined batch; 1.2 kg starting material.
Boundary and source	Fabrication steps only; fictional lab meter and material log. Upstream production excluded.
Personal goal	At most 10 kWh per valid sample, with unchanged quality and safety.

Confirm fields | Edit amounts or boundary | Mark unknown

One follow-up question: What makes a sample valid for this research? The proposed alternative can be compared only if it meets the same requirement.

Researcher
A valid sample passes our existing electrochemical test and safety checks. Keep those requirements for the pilot batch.

C. Review the initial profile and choose an action

Initial profile — before the pilot
The recorded batch used 60 kWh for 5 valid samples, or **12 kWh per valid sample**. The personal goal is **10 kWh per valid sample**; the initial gap is **2 kWh per valid sample above that goal**. Material loss and repeated attempts may be avoidable, but no improvement has yet been observed. Upstream material impacts remain unknown.

Suggested action: Pilot a process change that reduces failed samples; record electricity, starting material, valid samples and test results for the whole batch.

Plan pilot | Why this advice? | Correct input | Save privately

Criterion	Current evidence	Next check
C1 Boundary and inputs	Batch, validity rule, meter and material log recorded; upstream excluded.	Confirm the same steps and units in the pilot.
C2 Comparable alternative	Process change proposed; quality condition stated.	Test sample performance before claiming a fair comparison.
C3 Avoidable use	One failed attempt in the prior batch.	Check whether the change reduces failures or waste.
C4 Total use	Batch electricity and material recorded.	Show batch totals alongside per-valid-sample values.
C5 Follow-up	Personal goal recorded; result pending.	Set a review date and enter the pilot record.

Table 1: Illustrative assessment matrix using the five provisional criteria. These are evidence and action notes, not points to add into a score.

D. Optional return: compare the project's own history

The researcher later selects My History > Add follow-up and enters a fictional pilot batch: 6 attempts, 6 valid samples, 54 kWh and 1.1 kg starting material. The tool first checks that the scientific test, safety requirements and measurement boundary are unchanged. If they changed, it asks for a revised baseline instead of presenting the difference as an improvement.

Measure (same batch boundary)	Prior batch	Pilot batch	Change or goal gap
Valid samples / attempts	5 / 6	6 / 6	One more valid sample.
Batch electricity	60 kWh	54 kWh	−6 kWh for the batch.
Electricity / valid sample	12 kWh	9 kWh	−3 kWh from own baseline; −1 kWh against the 10 kWh goal.
Starting material	1.2 kg	1.1 kg	−0.1 kg for the batch.

Table 2: Fictional follow-up display. Differences are pilot minus the stated reference; negative electricity values indicate less use under the same boundary. A goal is a declaration, while the pilot values require recorded evidence.

Your project, so far: The pilot is 1 kWh per valid sample below your personal goal and 3 kWh below your own prior batch. Total batch electricity is 6 kWh lower. This is a project-specific observed change only if the meter record and quality checks are confirmed. It does not cover upstream impacts.

Optional similar-case view: If three reviewed cases genuinely match the functional unit, quality test and boundary, a *fictional demonstration mean* of 11 kWh per valid sample would place this pilot 2 kWh below that reference. The real interface would show the case count, range, sources and exclusions; otherwise it displays *no comparable reference available*. This comparison is context, not a rank.

View evidence | Revise goal | Query related cases | Return home

1.3 Option 2 — I want to Check & Learn: concepts and sources

This route implements the purpose described in Section 1.13: explain the tool, show why research activities matter, teach key concepts, and let researchers inspect a proposed Sustainability Library. Static cards and filters answer routine questions. AI is used only for a contextual explanation requested by the researcher. The library and its review roles are proposed, not deployed.

Home > 2. I want to Check & Learn

Our Story | Why It Matters | Key Concepts | Resources & Library | FAQ / User Guide

The researcher can open any card directly. A saved assessment is not loaded into this route unless the researcher explicitly selects it.

A. Short explanations and reusable concept cards

Selected: Our Story

This proposed tool helps TU/e researchers reflect on the environmental inputs of their own research activities, inspect the evidence behind advice, and decide what to change. It provides guidance rather than research approval.

Why It Matters | Start an assessment | Return to Learn menu

Selected: Why It Matters

Research activities can use electricity and materials and create waste. The relevant inputs depend on the activity, so the tool asks for a defined task and evidence before suggesting a change. It helps with reflection and does not approve or certify research.

See a battery example | Open Key Concepts | Return to Learn menu

Selected: Key Concepts

Functional unit	The result being compared, such as one sample that passes the same research test.
System boundary	The steps included in a quantity; a fabrication-only record excludes upstream material production.
Evidence status	A measured value, a sourced estimate and an unknown are labelled differently.

Apply to my assessment | Browse sources | Ask a question

B. Browse the proposed Sustainability Library

Selected: Resources & Library
Filter by source type: TU/e guidance | Scientific study | Case study | Framework | Database / external resource
Further filters: Activity | Scope | Date | Review status

The result list separates reviewed records from candidates awaiting expert review. Only reviewed, relevant records may support assessment advice.

Illustrative source card — institutional context
Title: TU/e Sustainability Visioning Report [Eindhoven University of Technology, 2023].
Origin and type: TU/e; institutional vision report.
Scope: Context for why sustainability matters at TU/e.
Limit: This card is not a battery-specific electricity or emissions factor.
Record fields before deployment: link, publication or update date, version, reviewer, review date and withdrawal status. Expert approval of this proposed library entry has not been established here.

View source details | Find activity method | Report a source issue

C. Ask for context only when the cards do not answer it

Researcher, after choosing "Ask a question"
My battery assessment shows upstream material impact as unknown. Why does the tool leave it blank?

AI-assisted explanation for this example
The confirmed activity boundary covers sample fabrication, and the upstream material production step was excluded. A number for that step would need suitable material quantities and a reviewed, matching method or factor. Until then, the result stays *unknown*; it is not zero. This explanation uses the activity boundary recorded in Assess. The TU/e vision report above provides institutional context, not a calculation input.

Review my boundary | Find a reviewed method | Return to Learn menu

FAQ / User Guide — fixed answers
Where did this advice come from? Open its source card to see origin, scope, date and review status.
Can I correct it? Use Correct assessment input or Report a source issue. A designated expert reviews source changes; AI does not add or approve official library entries.
Return home | Open Assess | Open Query

1.4 Option 3 — I want to Query: history and reviewed cases

The third home-menu choice supports two routes: a researcher's private history and a search of consented, reviewed cases. Fixed buttons handle navigation, filters and sharing choices; a short conversation is available when the researcher needs an explanation. The illustrative records below do not represent an existing TU/e database.

Home > 3. I want to Query

My History | Search reviewed cases | Guidance and methods | Return home

The following two paths illustrate the first two choices. A search result is never treated as a benchmark without checking its research purpose, quality requirement, unit and boundary.

A. My History: inspect a record or change sharing

Selected: My History

Your private assessment list

- **Case #2026-A:** Battery sample fabrication — baseline and follow-up recorded; case sharing enabled.
- **Case #2026-B:** Machine-learning training — draft; private.

Open A | Open B | Compare my versions | Manage sharing

Selected: Manage sharing > Case #2026-A > Private

Before confirming, the screen explains that the case will disappear from future shared searches, while previously downloaded copies cannot be recalled.

Confirm private | Cancel

On confirmation, the researcher's private history still contains both versions. Only a reviewed, consented and redacted case may appear in the shared search route.

B. Search: choose filters, inspect a case, challenge a source

Selected: Search reviewed cases

Activity: Battery testing | GPU computation | Other

Suggested filters: Same research purpose | Measured electricity | Matching boundary

The researcher selects battery testing and the three suggested filters. These are fixed controls populated from the confirmed activity; the researcher can change them.

Illustrative results: 12 reviewed battery-testing cases before filtering; three after the selected filters.

Case #TUe-BATT-88: cycle testing with grid electricity. Its shared view omits identifying details.
Matching check: equipment, test duration, valid outcome and measurement boundary still need inspection before a numerical comparison.

Open case details | Change filters | Return home

Case #TUe-BATT-88 — detail card

- Electricity: 420 kWh reported from a dedicated meter, with its test boundary shown.
- Upstream lithium supply: unknown; no zero value or footprint is inferred.
- A reported 12% change in estimated grid emissions requires its time-specific electricity factor, method and source before reuse.

Downloads: only approved data, units, boundaries and source versions. No overall sustainability score or public rank.

View source | Download approved data | Ask about a number

Researcher, after choosing "Ask about a number"
The carbon result has no emission factor or citation in the export. Can I use it in my lab report?

Assessment tool
That carbon result is unverified. The factor's value, unit, date, location and source must be supplied before the calculation can be checked or cited. I have flagged this record for curator review.

View available evidence | Report source issue | Return to results

1.5 Orientation and purpose

This section translates the Phase 1 ethical analysis into a testable web interface. A researcher should be able to see where a project stands, why an assessment was produced, what evidence is missing, and what action could improve it. The tool guides a structured reflection loop rather than assigning an unexplained sustainability score. Its intended output is an advisory, multidimensional profile with source-linked recommendations and a record of how the researcher's understanding or plan changes. It does not grant research approval. The retained Phase 1 report contains the detailed architecture and consent flow; the diagrams here focus on the user-facing prototype.

1.6 System structure and modules



Figure 1: Editable information hierarchy and navigation. Detailed processing and ownership appear below; the tree only maps the interface.

Module and purpose	Input	Processing and output	Why it matters
Home: orient	Goal choice.	Routes to three modules; states advisory scope.	Clear entry and limits.
Assess: revise	Project data; Start perception.	AI structures; formulas calculate; criteria support profile, advice and report.	Turns concern into inspectable decisions.
Check & Learn: understand	Question or concept.	Curated explanations, examples and sources.	Explains criteria and gaps.
Query: compare	Filters or consented history.	Retrieves reviewed cases with scope and conditions.	Supports valid comparison.

Table 3: Module contracts. The exact TU/e web entry point, authentication and maintainer role require client agreement.

1.7 Current gaps and improvement directions

TU/e has a sustainability vision and, according to the client Q&A documented in the retained Phase 1 process appendix, an existing researcher self-assessment. This prototype should complement that practice; its precise integration is unresolved. In a separate US survey of 64 graduate engineering researchers working with AI, 43 reported no environmental impact assessment of their work. Reported reasons included perceived irrelevance or small scale, lack of know-how and competing research priorities. These findings suggest design questions, not measured deficiencies among TU/e researchers [Shiekh and Zhu, 2026].

Element	Current status and gap	Evidence	Improvement direction
Awareness	TU/e has a vision and self-assessment; project-level impacts still need local testing.	Vision; client Q&A; [Shiekh and Zhu, 2026].	Ask for Start perception; show activity quantities; interview TU/e users.
Action	Information alone leaves no feasible next step.	[Shiekh and Zhu, 2026, Abrahamse et al., 2005].	Pair each gap with an action, constraint and source.
Explicability	Phase 1 calls for traceability; local factors and boundaries are unspecified.	[Friedman and Hendry, 2019].	Expose formula, source version, uncertainty and missing data.
Control and privacy	Correction and consent are proposed; reviewer and retention are unassigned.	[Santoni de Sio and van den Hoven, 2018, Nissenbaum, 2004].	Give correction and sharing controls; name a curator.
Comparable guidance	Relevant cases may suggest methods; their outcomes depend on activity and boundary.	[OECD and European Commission Joint Research Centre, 2008].	Show case conditions; track change against the researcher's own baseline.

Table 4: Gap analysis. The US survey motivates questions for local validation; it is not evidence of TU/e prevalence.

1.8 Example user journey and system processing

The example user is a researcher planning a six-hour experiment on four GPUs. They want to understand resource use before running it while preserving scientific quality. The journey inherits Phase 1's input, analysis, estimation, comparison, recommendation and feedback sequence.

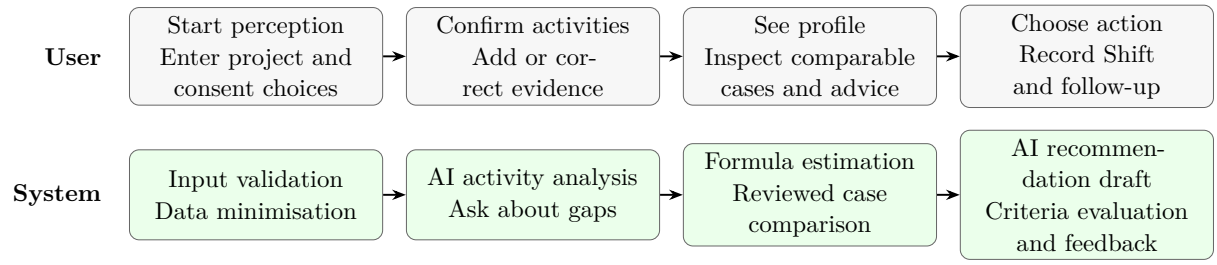


Figure 2: Two views of one workflow. User actions are above; internal processing is below. A correction returns to input validation and recalculation. The detailed optional case-sharing route remains in the retained Phase 1 workflow figure.

The system identifies activity types and missing fields, performs checkable calculations, retrieves

cases only when their boundaries are comparable, and drafts recommendations linked to reviewed sources. A researcher can reject a classification or revise an input; the system then recalculates and records what changed. Missing factors produce a qualified result rather than an invented estimate.

1.9 Criteria, profile and concrete output

The assessment follows the researcher's own activity rather than assigning a universal score. It records the scientific purpose, relevant resource use, data quality, a feasible change, and what happened after that change. Quantities retain their units and boundaries; missing information remains unknown. For example, four GPUs used for six hours amount to 24 GPU-hours, but this alone does not establish electricity use or emissions [Lacoste et al., 2019].

Table 5 turns five provisional criteria into prompts for a project-specific profile. A comparison is meaningful only when the research outcome, quality requirement, units and system boundary are comparable. A personal target is recorded as the researcher's intention; an observed improvement requires follow-up evidence.

Criterion	Project-specific input and evidence	Personal comparison and output
C1 Activity boundary and inputs	State the scientific outcome, functional unit, included steps, equipment or materials, amounts, duration, repeats and data sources.	Show relevant quantities with units and exclusions; ask for missing inputs.
C2 Comparable alternative	Describe the current method and a feasible alternative that meets the same research-quality requirement.	Compare with this project's baseline only where boundaries and evidence match; otherwise explain why comparison is pending.
C3 Avoidable use	Record failed or repeated runs, idle equipment or material loss, and a feasible action or pilot.	Identify a specific opportunity and action; record the result after testing.
C4 Wider impacts and total use	Include total runs or samples, other relevant energy, materials and waste, and any change in project scope.	Check whether a local saving shifts impacts or reduces total project use.
C5 Follow-up and revision	Keep the baseline, personal target, review date, comparable follow-up data and explanation of changes.	Show baseline, action and observed outcome separately; revise advice when evidence changes.

Table 5: Provisional activity-specific assessment prompts, adapted from the team's tutor-meeting outline. Measured values, sourced estimates and unknowns remain distinct. The criteria do not imply thresholds, weights, a ranking or an overall score.

Illustrative GPU-computation profile
Purpose and boundary: One successful run at the required scientific quality; also track all project runs.
Current record: Four GPUs for six hours (24 GPU-hours); electricity and emissions unknown without suitable power and facility data.
Personal aim: Test whether a smaller configuration can deliver the same result with less total electricity.
What is missing? Measured or defensible power, utilisation, failed runs and total planned runs.
Next action: Pilot a comparable configuration and record energy, run count and research quality.
Observed improvement: Pending follow-up; a projected saving is not presented as a measured result.
Controls: [Correct] [Challenge] [Save privately] [Review "Case A" for sharing].

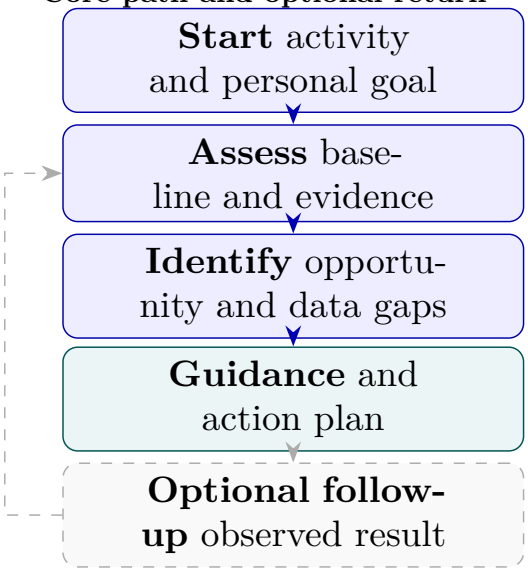
The output keeps environmental quantities, evidence quality and goal progress separate. The five criteria are prompts for tailored feedback, not a scale to add or average. A laboratory battery batch could instead track material losses and valid samples; a printed prototype could track failed prints and acceptable parts. In each case, advice must preserve scientific quality and safety. Cross-project examples may inform methods, but do not provide a ranking. Weighting unlike indicators into one number would require value judgements the group has not established [OECD and European Commission Joint Research Centre, 2008].

1.10 Start–Shift: project-specific progress

The researcher records an initial activity and a personal goal. The system identifies relevant quantities and evidence gaps, then suggests a feasible action. At a later visit, the researcher can enter comparable measurements and explain changes in research scope or quality. Figure 3 separates the first usable advice from optional follow-up; expected and observed outcomes remain distinct. Goal-reference research supports presenting change relative to a meaningful target: people evaluate gains and shortfalls relative to reference points, rather than only final totals [Kahneman and Tversky, 1979, Heath et al., 1999]. Formative-feedback work supports specific next steps rather than a terminal grade; self-regulation and personal-informatics models describe iterative, stage-based reflection and action [Shute, 2008, Zimmerman, 2002, Li et al., 2010]. These are design analogies, not evidence that this display will improve TU/e research behaviour. Comprehension, discouragement and actual follow-up must be tested [Carver and Scheier, 1982, Hattie and Timperley, 2007].

Consented cases from a proposed restricted repository could offer examples of feasible methods when activity and research-quality requirements match. This addresses the gap between general sustainability knowledge and locally relevant action illustrated by the graduate survey [Shiekh and Zhu, 2026]. Such examples would give context for the researcher's own decision, without ranking projects or researchers. TU/e could separately consider recognition for verified improvements, but awards or institutional incentives require their own governance.

Core path and optional return



Versioned phase matrix

Stage	Illustrative GPU-computation record
Baseline	Four GPUs for six hours; 24 GPU-hours
Evidence gap	Electricity, failed runs and total runs unknown
Goal	Same research quality with less total electricity
Action	Pilot fewer GPUs; measure comparable runs
Expected	Possible saving; no measured gain yet
Follow-up	Record actual energy, quality and total use

Figure 3: Solid blue arrows show the first usable result; muted dashed arrows mean a second or third assessment may occur, but is not required. Each relevant criterion can keep separate stage records, assumptions and source versions if the user chooses private storage. An estimate, a target and an observed improvement are different kinds of information; unknowns remain unknown.

1.11 Explicability, human control and privacy

The proposed knowledge base has three reviewed layers: authoritative TU/e guidance where applicable, published factors with scope and date, and consented cases whose conditions and outcomes have been checked. AI may classify activities, retrieve and summarise sources, propose comparators and draft suggestions. It may not create official policy, approve an archive entry or alter criteria without review. Fixed formulas perform arithmetic; an authorised maintainer validates source updates and can withdraw outdated advice. This separation makes outcomes traceable to both evidence and responsible people, reflecting the course's explicability and meaningful-human-control themes [Friedman and Hendry, 2019, Santoni de Sio and van den Hoven, 2018, European Commission, 2021].

The researcher chooses the detail entered, may omit unpublished or industry-confidential information, confirms AI-extracted activities, corrects assumptions, accepts or rejects advice, and chooses separately whether to save a private result, contribute to aggregate comparison, or share a reviewed case. Self-assessment remains available without archive sharing. The owner can recognise their own record; other users see only a consented, redacted case under a code (e.g., "Case A") or a sufficiently broad aggregate. A designated data steward would keep any identity–code mapping separately with restricted access; curators need only the case code and approved summary. Small groups and distinctive project details should be suppressed when they permit inference. A code is *pseudonymisation*, not guaranteed anonymity: linkable records remain personal data [European Data Protection Board, 2025, European Parliament and Council of the European Union, 2016]. This design applies purpose limitation, data minimisation and privacy by default, while leaving retention periods, legal basis and access roles for TU/e to determine [European Parliament and Council of the European Union, 2016, Nissenbaum, 2004]. A source challenge goes to a named reviewer and is not silently learned from as truth.

1.12 Evidence-to-design rationale

The source-to-design links are concentrated in Table 4. The US graduate survey motivates activity prompts and actionable guidance, but requires local testing [Shiekh and Zhu, 2026]. VSD makes explicability, autonomy and environmental impact design values to investigate with stakeholders rather than merely interface labels [Friedman and Hendry, 2019]. Feedback theory motivates Start–Shift, while meaningful human control and contextual privacy justify correction, curator review and separate sharing choices [Carver and Scheier, 1982, Hattie and Timperley, 2007, Santoni de Sio and van den Hoven, 2018, Nissenbaum, 2004]. OECD/JRC guidance supports keeping criteria visible and avoiding unsupported aggregation [OECD and European Commission Joint Research Centre, 2008].

1.13 Interface

The interface will be designed as a simple web-based tool with three main functions: I want to Assess, I want to Check & Learn, and I want to Query. The homepage will provide a clear overview of these functions so that researchers can choose between assessing their own research, learning more about sustainability, or exploring previous records and available resources.

The Assess function is the main AI-based part of the tool. The researcher first provides information about their research project, such as the type of research, activities, equipment, duration and other relevant details. The AI analyses this information and asks follow-up questions when important information is missing. It then identifies possible sustainability impacts and provides explanations and recommendations. The results can include aspects such as energy use, carbon emissions and other environmental impacts. The researcher can review the results, correct information or assumptions made by the AI, and then receive a final summary and guidance. The assessment is intended to support self-reflection and is not an official sustainability certification or approval.

The Check & Learn function provides the knowledge behind the assessment. It contains an introduction to the purpose of the tool, explanations of the connection between research and sustainability, key sustainability concepts, and a Sustainability Library. The library will contain guidelines, scientific literature, case studies, sustainability frameworks, relevant databases and external resources. Sources should come from reliable and reviewed sources, such as TU/e sustainability policies and guidelines, scientific publications, recognised sustainability organisations and relevant environmental databases. Each source can include information about its origin, publication or update date, scope and relevance to the assessment. This makes it possible for researchers to understand where the information comes from and to investigate the topic further.

The library will also contain examples of research activities and their possible sustainability impacts. These examples can help researchers understand how sustainability considerations can apply to different types of research. The AI can use the information from the reviewed knowledge base when analysing a research project and refer the researcher to the relevant sources. This means that the AI does not have to generate all sustainability information itself, but can use the library as a controlled source of knowledge. New sources can be added and existing sources can be reviewed or updated by designated sustainability experts. The AI itself should not independently add information to the official knowledge base.

The Query function allows researchers to explore information beyond their current assessment.

Researchers can view their own previous assessments and search the available database using filters such as research field, impact type or keywords. They can also explore relevant case studies, datasets, best practices and policies. When appropriate, comparable previous cases can help researchers understand how similar research activities have considered sustainability. Detailed information and relevant resources can be viewed or downloaded. Any sharing of research records should be subject to appropriate consent and privacy controls.

The interface will use clear headings, buttons and information cards. The three main functions will be visually separated so users can understand each purpose. Researcher input, AI-generated suggestions and retrieved records will be visually distinguished. The three walkthroughs are proposed screens for team review; Dennis will refine the Check & Learn content and its source presentation.

1.14 Evaluation plan and team responsibilities

The prototype will be compared with a structured non-AI checklist on the same test cases, as proposed in Phase 1. Measures are absolute calculation error, the share of recommendations supported by reviewed sources, the share of users who can explain a criterion, task completion and correction time, correct handling of sharing choices, and the share of suggestions judged feasible by domain reviewers. A separate test should compare a static profile with the profile-plus-gap display on comprehension, chosen action and later follow-up. Thresholds and reviewers must be agreed before testing; no result is claimed here.

Member	Lead responsibility	Shared work or handoff
Yehao Chen	Overall workflow, core concepts and shared navigation.	Assess steps and follow-up loop with Peng.
Dennis de Hoon	Check & Learn: Our Story, Why It Matters, Library and FAQ.	Explanations and source cards linked to Assess.
Peng Wang	Assessment criteria, interface and data presentation.	Assess steps and follow-up loop with Yehao.
Gerald Tobenna Eluagu	Query and history: records, search, filters and sharing controls.	Connect past records and reviewed cases to Assess.

Table 6: Current Phase 2 responsibilities and the shared Assess workflow.

References

Wokje Abrahamse, Linda Steg, Charles Vlek, and Talib Rothengatter. A review of intervention studies aimed at household energy conservation. *Journal of Environmental Psychology*, 25(3): 273–291, 2005. doi: 10.1016/j.jenvp.2005.08.002.

Charles S. Carver and Michael F. Scheier. Control theory: A useful conceptual framework for personality-social, clinical, and health psychology. *Psychological Bulletin*, 92(1):111–135, 1982. doi: 10.1037/0033-2909.92.1.111.

Eindhoven University of Technology. Tu/e sustainability visioning report. Technical report, TU/e Sustainability Ambassador Office, 2023. URL `https://assets.w3.tue.nl/w/fileadmin/TU_e_Sustainability/Sustainability_Team/Logos/TUe%20Sustainability%20Visioning%20Report%20Spread.pdf`.

European Commission. Ethics by design and ethics of use approaches for artificial intelligence. Technical report, Directorate-General for Research and Innovation, 2021. URL `https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ethics-by-design-and-ethics-of-use-approaches-for-artificial-intelligence_he_en.pdf`.

European Data Protection Board. Guidelines 01/2025 on pseudonymisation (version for public consultation), 2025. URL `https://www.edpb.europa.eu/system/files/2025-01/edpb_guidelines_202501_pseudonymisation_en.pdf`.

European Parliament and Council of the European Union. Regulation (eu) 2016/679 (general data protection regulation), 2016. URL `https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng`.

Batya Friedman and David G. Hendry. *Value Sensitive Design: Shaping Technology with Moral Imagination*. MIT Press, 2019.

John Hattie and Helen Timperley. The power of feedback. *Review of Educational Research*, 77 (1):81–112, 2007. doi: 10.3102/003465430298487.

Chip Heath, Richard P. Larrick, and George Wu. Goals as reference points. *Cognitive Psychology*, 38(1):79–109, 1999. doi: 10.1006/cogp.1998.0708.

Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979. doi: 10.2307/1914185.

Alexandre Lacoste, Alexandra Luccioni, Victor Schmidt, and Thomas Dandres. Quantifying the carbon emissions of machine learning. arXiv preprint arXiv:1910.09700, 2019.

Ian Li, Anind Dey, and Jodi Forlizzi. A stage-based model of personal informatics systems. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pages 557–566. ACM, 2010. doi: 10.1145/1753326.1753409.

Helen Nissenbaum. Privacy as contextual integrity. *Washington Law Review*, 79(1):119–157, 2004. URL `https://digitalcommons.law.uw.edu/wlr/vol79/iss1/10/`.

OECD and European Commission Joint Research Centre. *Handbook on Constructing Composite Indicators: Methodology and User Guide*. OECD Publishing, Paris, 2008. doi: 10.1787/9789264043466-en.

Filippo Santoni de Sio and Jeroen van den Hoven. Meaningful human control over autonomous systems: A philosophical account. *Frontiers in Robotics and AI*, 5:15, 2018. doi: 10.3389/frobt.2018.00015.

Kylee Shiekh and Qin Zhu. Exploring how graduate engineering researchers approach AI sustainability: Perceptions, design considerations, and impact assessment. *International Journal of Technology and Design Education*, 2026. doi: 10.1007/s10798-026-10100-y. URL `https://link.springer.com/article/10.1007/s10798-026-10100-y`.

Valerie J. Shute. Focus on formative feedback. *Review of Educational Research*, 78(1):153–189, 2008. doi: 10.3102/0034654307313795.

Barry J. Zimmerman. Becoming a self-regulated learner: An overview. *Theory Into Practice*, 41 (2):64–70, 2002. doi: 10.1207/s15430421tip4102_2.